

CASE REPORT | Liver

원발성 위 미만 거대 B세포 림프종의 자발적 소실에 대한 드문 증례 보고

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Rare Case of Spontaneous Regression in Primary Gastric Diffuse Large B Cell Lymphoma

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The spontaneous regression (SR) of primary gastric diffuse large B-cell lymphoma (PG-DLBCL) is extremely rare. Although the mechanism of SR is unclear, the host immune response, ischemia, or changes in the tumor microenvironment may play a role. On the other hand, SR does not indicate a complete cure and residual lymphoma may remain dormant, leading to relapse. This paper presents a rare case of an 82-year-old woman with PG-DLBCL who experienced SR without treatment, followed by a short-term relapse. In the present case, the most likely mechanism was biopsy-induced ischemia and increased apoptosis. This case underscores that, in PG-DLBCL with SR, the initial endoscopic findings are the most critical reference for an accurate diagnosis and proper management, but careful long-term follow-up and monitoring are necessary, given the risk of relapse. (*Korean J Gastroenterol* 2025;85:221-225)

Key Words: Lymphoma, large B-cell, diffuse; Regression, spontaneous

INTRODUCTION

Spontaneous regression or remission (SR) of a malignant tumor is very rare and is defined as the partial or complete disappearance of a tumor without any medical treatment or therapy known to have antitumor effects.^{1,2} This phenomenon has been documented in several tumors, including neuroblastomas, testicular tumors, renal cell carcinomas, melanomas, breast cancers, and lymphomas.²⁻⁵ Although SR remains an enigmatic and poorly understood condition, its clinical im-

plications for diagnosis and treatment are important because there is a significant risk of relapse.^{1,6} In the context of primary gastric malignancy, early gastric cancer (EGC) may show endoscopic improvement, particularly after proton pump inhibitor or *Helicobacter pylori* (*H. pylori*) eradication therapy.⁷⁻⁹ Nevertheless, SR in the setting of a diffuse large B cell lymphoma (PG-DLBCL) is exceedingly rare, with only a few documented cases in the literature.

This paper presents a rare case of PG-DLBCL that demonstrated SR in the absence of any specific treatment followed

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by a short-term relapse. This case highlights the exceptional nature of SR in PG-DLBCL and raises the importance of close monitoring in patients suspected of SR.

This study was approved by the Institutional Review Board of Pusan National University Hospital (No. 2411-004-144).

CASE REPORT

An 82-year-old woman was referred to the authors' gastroenterology clinic for the further evaluation of a superficial gastric ulcer with irregular margins detected by screening esophagogastroduodenoscopy (EGD) performed two months before the visit (Fig. 1A). The patient was asymptomatic, with no gastrointestinal complaints, and her physical examination was

unremarkable. Although the initial forceps biopsy specimen revealed non-specific inflammatory cells in the lamina propria compatible with chronic gastritis (Fig. 1B), the endoscopic findings were highly suspicious of an EGC or gastric lymphoma. A second EGD was performed promptly, and no prescriptions were given before the examination. Upon repeat EGD, the previous gastric ulcer appeared to have almost healed, leaving only a small raised erosion with a scar on the posterior wall of the proximal antrum (Fig. 2A). A follow-up biopsy of the remaining erosive lesion revealed the focal presence of atypical lymphoid cells with a squeezing artifact that was positive for CD20, but negative for CD3 and pancytokeratin on immunostaining (Fig. 2B–E). The insufficient specimen volume limited the authors' capacity to confirm a definitive diagnosis,

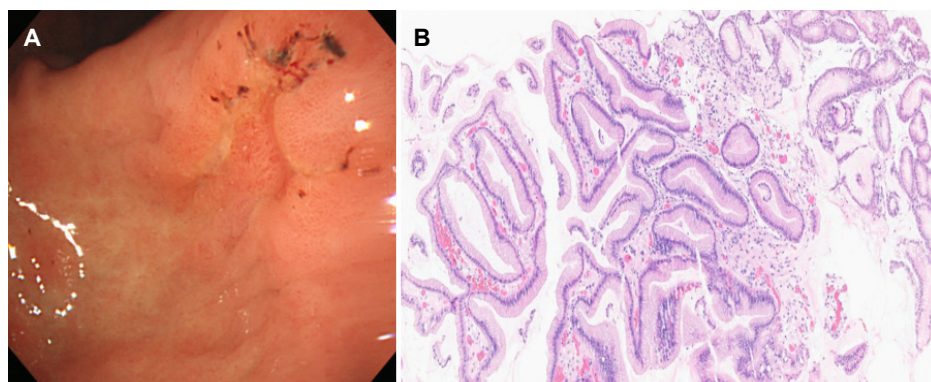


Fig. 1. (A) On initial screening endoscopy, a superficial gastric ulcer with irregular margins was observed on the posterior wall of the proximal antrum. (B) The initial biopsy specimen shows non-specific inflammatory cells, and there is no evidence of malignancy (H&E stain, $\times 100$).

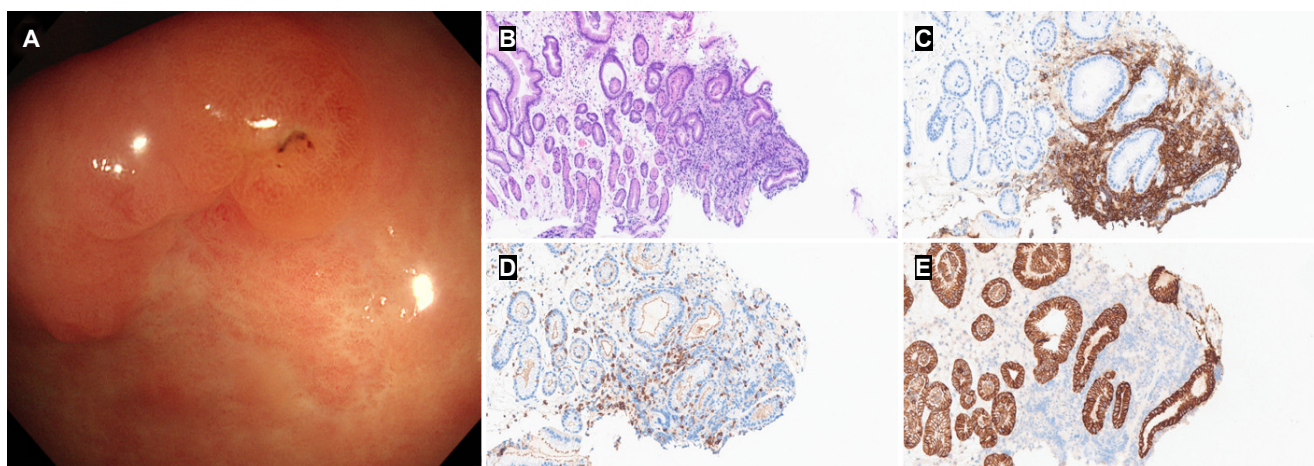


Fig. 2. (A) Two months later, a follow-up esophagogastroduodenoscopy revealed that the previously noted gastric ulcer had nearly healed, leaving only a small raised erosion with scarring on the posterior wall of the proximal antrum. (B) The second biopsy specimen displays some atypical lymphoid cells at the edge of the specimen that are worrisome for neoplastic conditions (H&E stain, $\times 200$). The immunohistochemical findings showed that these atypical cells were positive for CD20 (C), negative for CD3 (D), and pancytokeratin (E) (magnification, $\times 200$).

but the morphological and immunohistochemical findings suggested B-cell lineage lymphoma. The next EGD was planned to obtain a definite diagnosis, but the procedure was delayed

and conducted six months later due to the patient's circumstances. The third EGD showed that the previous erosion had completely healed (Fig. 3A). On the other hand, three

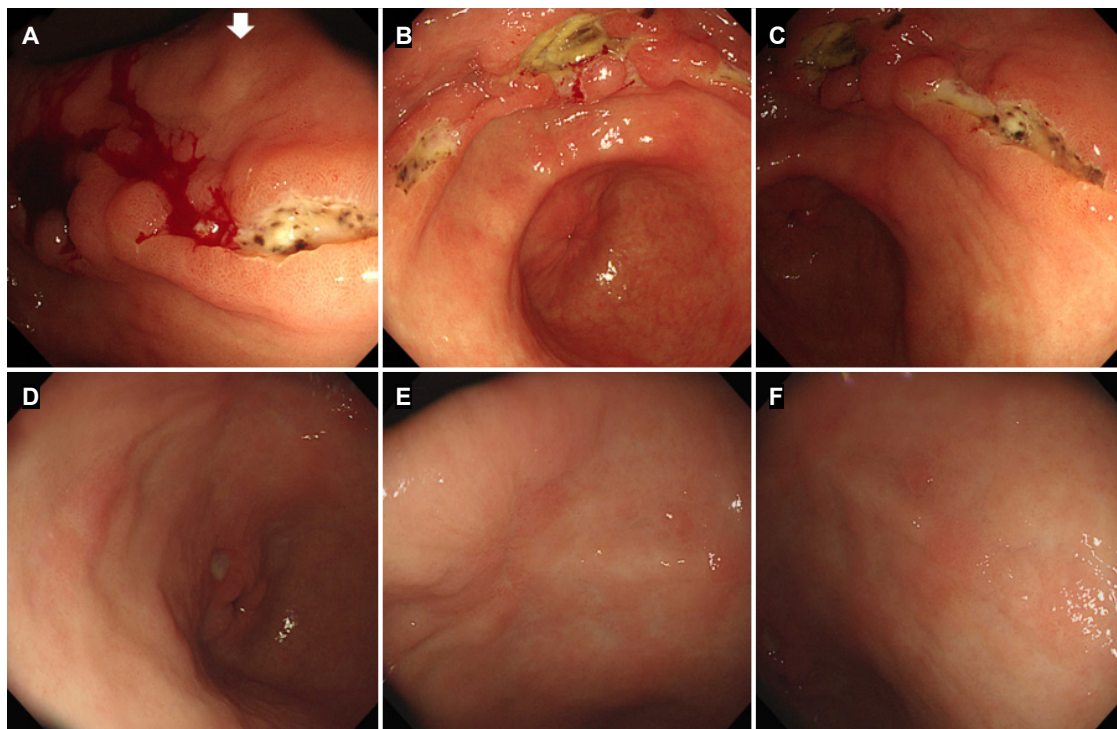


Fig. 3. (A–C) Eight months after the initial endoscopy, the previous gastric ulcer had completely healed (white arrow), but three newly developed gastric ulcers with a heaped-up margin covered by yellow to white exudate were observed on the proximal antrum along the lesser curvature. (D–F) After six cycles of chemotherapy, the lymphoma-related ulcers resolved completely, leaving behind fibrotic scars.

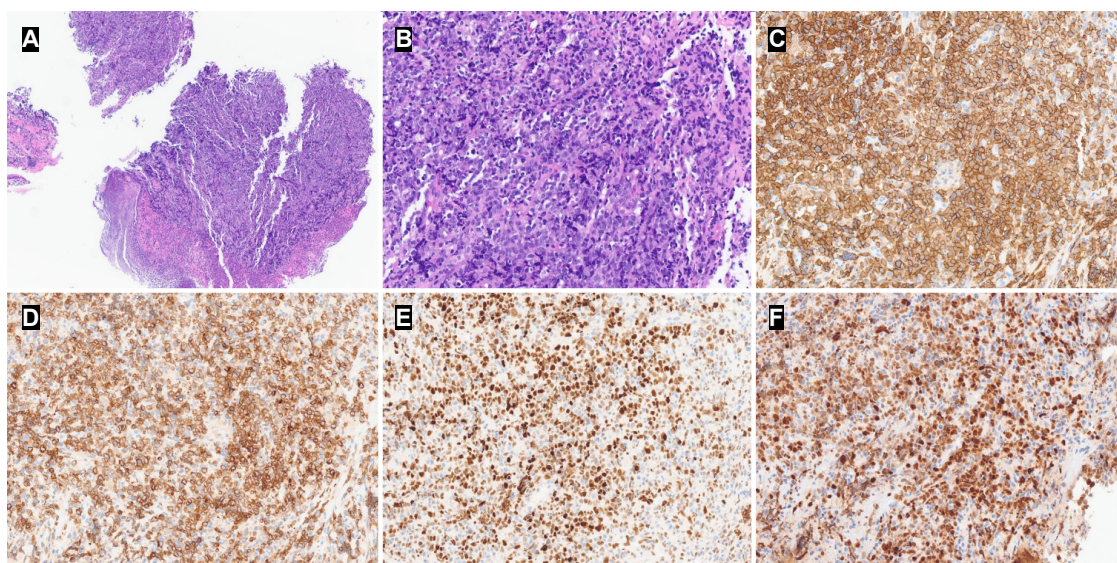


Fig. 4. (A) Normal gastric tissue architecture was effaced by tumor cells (H&E stain, ×40). (B) Large-sized atypical lymphoid cells diffusely infiltrate the lamina propria (H&E stain, ×200). The immunohistochemical findings showed that the tumor cells were positive for CD20 (C), BCL2 (D), BCL6 (E), and MUM1 (F) (magnification, ×200).

newly developed gastric ulcers with heaped-up margins covered by yellow to white exudate were observed on the proximal antrum along the lesser curvature (Fig. 3A–C). The endoscopic morphology raised strong suspicion for DLBCL, particularly because of the characteristic auricular sign. The biopsied specimen revealed the diffuse infiltration of large-sized atypical lymphoid cells with prominent nuclei and vesicular chromatin and the normal tissue architecture effaced by tumor cells. Immunohistochemistry revealed tumor cells positive for CD20, BCL2, BCL6, and MUM1 (Fig. 4), but negative for CD3, CD5, CD10, and pancytokeratin. EBER in situ hybridization was also negative. The morphological and immunohistochemical findings were consistent with a diffuse large B-cell lymphoma, a non-germinal center B-cell type. Lymphoma staging showed no involvement of other organs or lymph nodes, confirming the diagnosis of PG-DLBCL. The lymphoma-related ulcers completely healed after the R-CHOP chemotherapy (Rituximab, Cyclophosphamide, Doxorubicin Hydroxydaunomycin, Vincristine, and Prednisone) for six cycles (Fig. 3D–F). A follow-up biopsy showed no lymphoma cells and was considered complete remission.

DISCUSSION

PG-DLBCL is a rare and aggressive malignancy that usually requires aggressive treatment strategies, including chemotherapy and radiotherapy.¹⁰ Although the prognosis of PG-DLBCL is typically poor without proper treatment, there are exceptional cases in which PG-DLBCL undergoes SR.^{6,11} Although SR in PG-DLBCL is extremely rare, and the precise mechanism is unclear, several potential hypotheses have been suggested.^{1,4,6,11} An enhanced host immune response is one of the most important proposed mechanisms.^{1,4} Lymphoma, particularly a B-cell malignancy, is closely associated with immune surveillance and response, and the enhanced immune response through the activation of natural killer cells and cytotoxic T lymphocytes (CD8+ T cells) may have mounted a sufficient response to eradicate the tumor.^{4,12,13} Factors that stimulate an immune response, including fever and infection, may also promote SR by activating tumor cells to release cytokines.⁴ In some reports, chronic infections, particularly *H. pylori* have been implicated in the SR of lymphoma through a local immune activation, and *H. pylori* eradication could lead to tumor regression by modifying

the tumor microenvironment.^{1,11,14} Some cases of SR after taking an H2 receptor antagonist (H2RA) have been reported. These studies suggested that H2RA might have an immunomodulatory effect and alter the tumor microenvironment, but the evidence is limited.^{11,15} Furthermore, SR can occur if the tumor becomes hypoxic or ischemic, leading to reduced blood flow, and inducing increased apoptosis.¹ In the present case, the patient did not take any medications other than those for diabetes mellitus and hypertension, and there was no evidence of infectious illness before SR. Although it is difficult to determine the precise mechanism of SR, biopsy-induced ischemia and increased apoptosis is believed to be the most likely mechanism in the present case. The initial lesion might not have been DLBCL but rather a B-cell lineage mucosa-associated lymphoid tissue lymphoma (MALT) lymphoma. Nevertheless, it is more likely that the initial lesion was DLBCL that underwent SR considering that rapid transformation from MALT lymphoma to DLBCL is extremely rare.¹⁶

SR is a remarkable event, but this does not mean the lymphoma in the present patient had been completely cured. In many cases of SR, the regression may not involve the destruction of the tumor but rather a reduction of tumor cells or disappearance of visible tumor masses (minimal residual disease).^{1,17} Residual lymphoma cells may remain in a latent form, and they can reactivate under certain conditions where the tumor microenvironment changes.¹⁸ Relapse can occur months or even years after the initial SR. Therefore, in clinical practice, long-term follow-up and careful monitoring are necessary in SR cases.^{1,6}

In the present case, there are some important clinical implications to consider. When SR occurs in PG-DLBCL, the regressed lesions might show signs of healing or scarring, with flattened mucosa, as shown in the present case. This can lead to a potential diagnostic challenge, as SR can mimic other common benign gastric conditions such as gastritis, erosion, and ulcers. However, it is crucial to consider that gastric malignancies, including PG-DLBCL, can sometimes regress spontaneously without treatment. Even if specific lesions that appear malignant show endoscopic improvement, this does not necessarily exclude a malignancy. In this situation, the initial endoscopic findings serve as a critical reference point, determining the appropriate clinical approach for an accurate diagnosis.

In conclusion, this paper reported a rare case of SR in

PG-DLBCL followed by a short-term relapse. A thorough understanding of the possibility of SR in PG-DLBCL is essential, and the initial endoscopic findings are the most critical reference for an accurate diagnosis and proper management of PG-DLBCL. In cases when SR is suspected, careful long-term follow-up is necessary, considering the risk of relapse.

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